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Engineering a Deception: What Led to Volkswagen's Diesel Scandal

By JACK EWING MARCH 16, 2017

Photo



Credit Sean Gallup/Getty Images

In September 2015, Volkswagen was [accused of installing illegal software](#) on cars to evade standards on diesel emissions.

Revelations of the deception led to the resignation of the company's chief executive and have hit the company's share price and profits.

Here is a timeline of how the scandal unfolded, according to a plea agreement by the carmaker in the United States and other court documents, as well as interviews with some of the participants.

The Planning

2006

Volkswagen wants to increase its embarrassingly low market share in the United States as part of a plan to become the biggest car

Nov. 2006	company in the world. To do this, it resolves to use fuel-efficient diesel engines as a way to compete with Toyota. But engineers realize a newly developed diesel motor cannot meet stricter American emission standards.
The Deception	
Late 2008	At a contentious meeting at company headquarters in Wolfsburg, Germany, executives decide to install illegal software that can detect when an emissions test is taking place and crank up pollution controls, according to two people who saw a PowerPoint presentation laying out details of the plan and spoke on condition of anonymity. The rest of the time the controls are throttled back to protect engine components from wear and tear.
Late 2011	The Volkswagen begins a “clean diesel” marketing push in the United States, positioning its Golfs, Jettas, Beetles and other models as environmentally friendly alternatives to hybrids like the Toyota Prius.
	Volkswagen begins selling Passat sedans made at a new, \$1 billion factory in Chattanooga, Tenn. Though the cars are equipped with more advanced emissions technology, Volkswagen later admits in court documents that it continued to use cheating software to reduce the amount of customer-level maintenance required.
Investigation	
2013	A team from West Virginia University wins a \$70,000 grant from the International Council on Clean Transportation to test whether diesel cars produce more emissions during normal driving than during laboratory tests. Road tests reveal that two Volkswagens in the study emit far more poisonous nitrogen oxides than allowed.
May 2014	
Spring 2014	After publication of the West Virginia study , a Volkswagen product-troubleshooter warns top managers in a memo that regulators may investigate whether the carmaker has been deploying an illegal “defeat device.” The company confirms that the troubleshooter’s memo was passed on to Martin Winterkorn, then Volkswagen’s chief executive, but Volkswagen later says that he may not have read it.
Spring 2015	The California Air Resources Board, or C.A.R.B., opens an inquiry into the excess Volkswagen emissions uncovered by the West Virginia study. Volkswagen executives and engineers try to cover up the illegal software by feeding the regulators false and misleading data, the company later admits in its plea agreement.
The Revelation	
Aug. 2015	Volkswagen recalls diesel vehicles in the United States, telling regulators that a software update will fix the excess emissions. After tests show only a modest improvement, C.A.R.B. intensifies its inquiry.
Sept. 3, 2015	As regulators ramp up the pressure, Volkswagen employees destroy

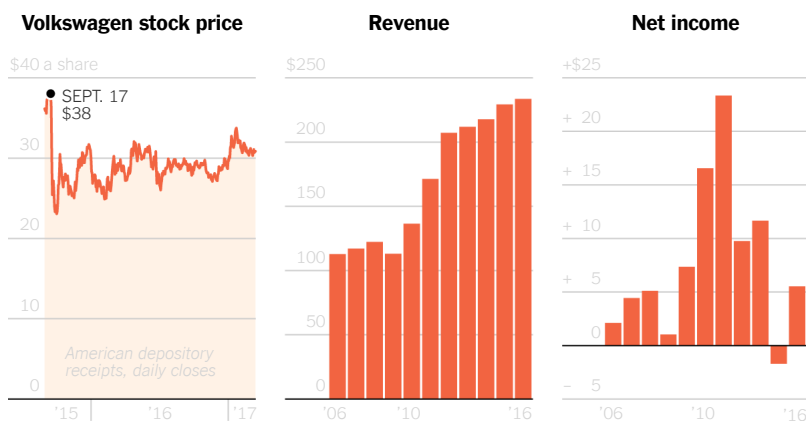
The Fallout

thousands of potentially incriminating documents, the carmaker admits in court documents.

Sept. 19, 2015

Volkswagen confesses to the Environmental Protection Agency and C.A.R.B. that its diesel cars are equipped with defeat-device software that detects when emissions tests are taking place and cranks up pollution controls so that the cars will pass.

The public learns of the fraud after the E.P.A. gives Volkswagen a [formal notice of violation](#).



Sept. 23, 2015

[Mr. Winterkorn resigns](#), but says he is not aware of any wrongdoing on his part. Volkswagen insists that the illegal software was the work of a small group of rogue engineers.

June 2016

Sept. 2016

Volkswagen [agrees to a \\$14.7 billion settlement](#) in the United States to deal with civil complaints and consumer lawsuits. The settlement includes compensation to owners and measures to atone for the cheating, such as installing networks of electric car charging stations.

Jan. 2017

A longtime Volkswagen engineer [pleads guilty](#) in the United States to conspiring to defraud regulators and car owners, the first criminal charges stemming from the American investigation into the automaker's emissions deception.

Graphic by Karl Russell

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March 2017

In a separate settlement with the United States Justice Department, [Volkswagen agrees to plead guilty](#) to conspiring to defraud the United States government and violate the Clean Air Act. The company will pay a \$4.3 billion fine. The formal plea is [submitted in March](#).

[United States prosecutors separately indict](#) six former Volkswagen executives, including an engineer arrested at Miami International Airport.

Volkswagen [agrees to pay](#) a further \$1.3 billion to owners of Audi, Porsche and Volkswagen vehicles with three-liter motors not covered by the earlier settlement. The settlement brings the total cost to Volkswagen of the scandal in the United States to more than \$21 billion.



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The auto components maker [Robert Bosch agrees to pay](#) \$327.5 million to vehicle owners as compensation for its role in supplying the cheating software, but it does not concede wrongdoing.

[German prosecutors say](#) that 37 people are suspected of taking part in the fraud, including Mr. Winterkorn.

German prosecutors [raid offices of Audi](#), Volkswagen's luxury car unit and a major profit center, in locations around Germany, escalating the investigation into the company's wrongdoing in the United States.

Matthias Müller, the Volkswagen chief executive, says at a [news conference](#) that the fraud was the work of lower-ranking employees and did not include any members of the current management board.

[VW Pleads Guilty to Criminal Felony in Diesel Scandal](#)

[Dec. 22, 2017](#)



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